PROGRAMME : Diploma Programme in IF / CM
COURSE : Information Security(IFS) **COURSECODE :** 232404

LEARNING AND ASSESSMENT SCHEME:

Learning Scheme					Credits	Assessment Scheme											Total Marks
Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL and TSL Practical				Based on self Learning		
CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
							Max	Max	Max	Min	Max	Min	Max	Min	Max	Min	
03	00	02	01	06	03	03	30	70	100	40	25	10	-	-	25	10	150

IKS Content : Total Learning Hours for Term : 00 Hrs
# Indicates Online Examination, @ Internal assessment

1.0 RATIONALE:

The goal of Information Security is to familiarize students with the security issues and technologies involved in modern information systems. Students will gain an understanding of the various ways in which information systems can be attacked and tradeoffs in protecting networks. Students will gain an appreciation of the need to develop an understanding of underlying system applications and potential security issues early in the design process.

New communication systems and digital technology have made dramatic changes in the way we live and the means to transact our daily business. Businessmen are increasingly using computers to create, transmit and store information in electronic form instead of traditional paper documents. It is cheaper, easier to store and retrieve and speedier to communicate. This will enable them to develop a sound knowledge and analytical ability facilitating their intellectual and professional development and future employment

INDUSTRY / EMPLOYER EXPECTED OUTCOME:

The aim of the course is to help the student to attain the following industry /employer expected outcome:

Apply network security threads, methods of data security and authentication .

2.0 COURSE OUTCOMES:

The course content should be taught and learning imparted in such a manner that students are able to acquire required learning outcome in cognitive, psychomotor and affective domain to demonstrate following course outcomes:

1. Describe pillars of information security and principles of information security
2. Understand and apply the cryptography technique
3. Understand and apply the different web security techniques.
4. Demonstration of recovery of data from different ways
5. Overview of different standards and frameworks

3.0 COURSE DETAILS:

Unit	Major Learning Outcomes (in cognitive domain)	Topics and Sub-topics	Hours
Unit-I **Introduction to Information Security architecture and model**	1a. Recall basic of Information 1b. Describe information classification 1c. Analyze information security management 1d. State and explain security policies and standard.	1.1 Information, Need and Importance of Information, information classification, criteria for information classification 1.2 Security, need of security, Basics principles of information security	07

Unit	Major Learning Outcomes (in cognitive domain)	Topics and Sub-topics	Hours
	1e. Define concept of TCB 1f. Comprehend common criteria for information security evaluation	1.3 Information security and Risk Management, Security policies, guidelines, standards 1.4 Trusted computing base, Rings of Trust, Protection Mechanisms in a trusted Computing Base 1.5 System security assurance concepts, Trusted computer security Evaluation Criteria 1.6 Information Technology security Evaluation Criteria, Confidentiality and Integrity Models.	
Unit-II **Cryptography**	2a. Describe Common term related to field of cryptography 2b. Demonstrate how to encrypt and decrypt messages using transposition and substitution method 2c. Define stenography 2d. Purposes and uses of digital signatures	2.1 Introduction, Application of cryptography, Classical encryption Techniques, Symmetric cipher 2.2 Substitution cipher • Ceasor cipher • Playfair cipher • Hill cipher 2.3 Transposition cipher • Row transposition cipher • One Time Pad 2.4 Steganography, Digital Signatures, Authentication Protocols, Digital Signature Standards	10
Unit-III **Application and web security**	3a. Understand web security 3b. Identify various security techniques for web security	3.1 Application hardening, application patches, web servers, active directory 3.2 Web security threats, web traffic security approaches, secure socket layer and transport layer security, secure electronic transaction 3.3 Software development: secure code techniques, buffer overflows, code injection, least privilege, good practices, requirements, testing	10
Unit-IV **Data Recovery and cyber security**	4a. Understand recovery of data from different ways 4b. Remember about different cyber crimes 4c. Know IT acts in India	4.1 Introduction to Deleted File Recovery ,Formatted Partition Recovery, Data Recovery Tools, Data Recovery Procedures and Ethics. 4.2 Introduction to Cyber Crimes – Hacking, Cracking, Viruses, Virus Attacks, Pornography, Software Piracy, Intellectual property, Legal System of Information Technology, Mail	10

Unit	Major Learning Outcomes (in cognitive domain)	Topics and Sub-topics	Hours
		Bombs, Bug Exploits, Cyber Crime Investigation 4.3 Introduction Cyber Laws- Introduction to IT act 2000 and IT act 2008, Introduction to the cyber laws	
Unit-V **Access, physical control and compliance standards**	5a. State the need of physical security 5b. Differentiate different Authorization and Authentication mechanism 5c. Overview of different standards and frameworks	5.1 Identification, Authorization, Authentication, Biometrics, Single Sign –on, Kerberos, Remote user access and Authentication. 5.2 Physical access control, Physical access threats, providing physical security. 5.3 Compliance standards: Implementing and Information Security Management System, ISO 27001, ISO 20000, BS 25999, PCI DSS, ITIL framework, COBIT framework	08
		TOTAL	**45**

4.0 SUGGESTED SPECIFICATION TABLE WITH MARKS (THEORY):

Unit No.	Unit Title	Distribution of Theory Marks			
		R Level	**U Level**	**A and above Levels**	**Total Marks**
I	Introduction to Information Security Information architecture and model	06	06	04	16
II	Cryptography	06	04	06	16
III	Application and web security	04	04	04	12
IV	Data Recovery and cyber security	04	06	04	14
V	Access, physical control and compliance standards	04	04	04	12
	TOTAL	**24**	**24**	**22**	**70**

Legends: R = Remembrance (Knowledge); U= Understanding; A= Application and above levels (Revised Bloom's taxonomy)

Note: This specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary slightly from above table.

5.0 LABORATORY LEARNING OUTCOME AND ALLIED PRACTICAL/ TUTORIAL EXPERIENCES:

The tutorial/practical/assignment/task should be properly designed and implemented with an attempt to develop different types of cognitive and practical skills **(Outcomes in cognitive, psychomotor and affective domain)** so that students are able to acquire the desired programme outcome/course outcome.

Sr. No.	Unit No	Learning Outcomes	Practical Exercises (Outcomes in Psychomotor Domain)	Approx. Hrs. required
1	I	Understand the security of operating system	Knowing the security provided with windows operating system	02
2	II	Implement recovery of password.	Recovery the password of windows machines using password recover utility (John the	02

Sr. No.	Unit No	Learning Outcomes	Practical Exercises (Outcomes in Psychomotor Domain)	Approx. Hrs. required
			ripper) or any other utility	
3	II	Implement tracing of email	Tracing of email origin using eMailTracePro utility	02
4	II	Use Key logger.	Use of keylogger and anti-key logger to secure your system	02
5	II	Understand Encryption and Decryption.	Encrypt and decrypt the message using Simple Transposition –Permutation(Cryptool)	04
6	II	Implement Caesar cipher.	Encrypt and decrypt the message using Caesar Cipher With Variable	04
7	III	Implement Hill Cipher.	Encrypt and decrypt the message using 3 X 3 Hill Cipher(Cryptool)	04
8	IV	Create digital signature	Create Digital Signature document (using Cryptool)	04
9	IV	Understand steganography technique.	Send and receive secret message using Steganography techniques using steghide	02
10	IV	Understand pen drive and hard disk	Recover the data from formatted Pen drive and Hard Disk using power data Recovery utility or any other utility	04
			TOTAL	**30**

Note: Here only outcomes in psychomotor domain are listed as practical/exercises. However, if these practical/exercises are completed appropriately, they would also lead to development of **Programme Outcomes/Course Outcomes in Affective domain** as given in the mapping matrix for this course. Faculty should ensure that students also acquire Programme Outcomes/Course Outcomes related to affective domain.

Minimum 10 or more practical's need to be performed out of which the practical's marked as '*' are compulsory.

6.0 SELF LEARNING: (Assignment / Activities for Specific Learning / Skill Development / Online Courses / Micro projects

Other than the classroom and laboratory learning, following are the suggested Student-related co-curricular activities. This can be undertaken to accelerate the attainment of the various outcomes in the course,

Suggested Self Learning Activities Following are some suggestive self-learning topics:

1. Prepare charts on cyber security laws and acts.
2. Prepare charts on security policies and standards.
3. Case study on cyber security challenges
4. Assign privileges to various users for checking access control over systems and networks
5. Apply authentication techniques for securing data

7.0 SPECIAL INSTRUCTIONAL STRATEGIES (If any):

These are sample strategies, which the teacher can use to accelerate the attainment of the various outcomes in this course,

1. Use different types of teaching methods and media to develop the outcomes.
2. For self learning activity teacher needs to create the opportunity and provision for co-curricular activity.

3. Guide students to undertaking mini-project
4. Encourage students to refer different websites to deeper understanding of subject.
5. Visit cyber investigation cell to know more about cyber laws
6. Arrange Expert Lectures on Ethical Hacking
7. Observe continuously and monitor the performance of students in Lab.

8.0 ASSESSMENT METHODOLOGY

Formative assessment (Assessment for Learning)

Lab performance, Assignment, Self-learning.

TH – 02 Progressive/Periodic test each of 30 marks.

TL – Continuous assessment of tutorials for 25 marks.

Self learning assessment 25 marks.

Summative Assessment

TH – End examination of 70 marks

9.0 LEARNING RESOURCES:

A) Books:

Sr. No.	Title of Book	Author	Publication
1	Principles of Information Security	Whitman	Cengage India
2	Information System Security	Godbole Nina	John Wiley
3	Information Security Principles and Practices	Mark Merkov & Jim Breithaupt	Pearson
4	Cryptography and Information Security	V.K. Pachghare	Prentice Hall India
5	Information Security and Cyber Laws	Saurabh Sharma	Vikas Publishing House
6	Computer Networking	Tularam M. Bansod	Dreamtech Press

B) Software/Learning Websites:

a. CrypTool 1.4.21 (www.cryptool.org)
b. http://www.emailtrackerpro.com
c. http://www.kmint21.com (Keylogger)
d. http://www.jjtc.com/Steganography/tools.html
e. http://steghide.sourceforge.net/ (stegnography)
f. http://www.powerdatarecovery.com/ (data recovery)

C) Major Equipment/ Instrument with Broad Specifications:

1. Computers for language laboratory software
2. Headphones with microphone.
3. LCD projector.
4. Open source tools -: Cryptool

10.0 MAPPING MATRIX OF PO's, CO's and PSO's:

H: High Relationship, M: Moderate Relationship, L: Low Relationship.

Course Outcomes	Programme Outcomes (PO's)							Programme Specific Outcomes (PSO's)			
	1	2	3	4	5	6	7	1	2	3	4
CO1	H	M	L	H	M	L	M	H	L		
CO2	H	M	H	L	M	L	H		H		
CO3	H	L	M	H	L	M	M		L		
CO4	M	M	H	L	L	H	L			L	L
CO5	L	M	H	M	H	M	M			L	L